Table 6: OLS Regressions. Winning Rate for Refusals

	Ethical Refusals			Technical Refusals		
Variable	**Win**	**Loss**	**Tie**	**Win**	**Loss**	**Tie**
Similarity	0.016***	-0.023***	0.006	0.025***	-0.033***	0.008
	(0.006)	(0.008)	(0.008)	(0.006)	(0.008)	(0.008)
Similarity Opp.	-0.011*	0.036***	-0.025***	-0.047***	0.037***	-0.010
	(0.006)	(0.008)	(0.008)	(0.007)	(0.009)	(0.009)
Length	0.090***	-0.062**	-0.027	0.026	-0.025	-0.001
	(0.029)	(0.03)	(0.035)	(0.030)	(0.034)	(0.036)
Length Opp.	-0.009	0.046***	-0.037***	-0.012	0.066***	-0.053***
	(0.006)	(0.011)	(0.010)	(0.010)	(0.014)	(0.012)
Controls LLM Model	Yes	Yes	Yes	Yes	Yes	Yes
Controls Opp. Model	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.11	0.51	0.42	0.10	0.22	0.15
Observations	2655	2655	2655	3202	3202	3202

Robust standard errors in parenthesis.* p<.1, ** p<.05, ***p<.01. Controls LLM includes dummy variables for the LLM used for the response and of the opponent model. Controls Opp. Model controls for the response category of the Opponent Model

To assess whether the effects observed for ethical refusals generalize to technically motivated refusals, we replicate the analysis for Refusal Technical responses (Table 6 columns 4 – 6). The results reveal a similar pattern for textual similarity, which is again positively associated with win rates and negatively with loss rates. In fact, the magnitude of these effects is slightly stronger than for ethical refusals. A one standard deviation increase in similarity corresponds to a 2.5 percentage point increase in win rates for user decisions. However, we do not find significant effects of text length for technical refusals.

Taken together, we find support for H3a and partial support for H3b—textual similarity improves evaluations across refusal types, while response length only has a significant effect for ethical refusals.

4.3. Prompt Sensitivity and User Reactions to Ethical Refusals

The reception of refusal responses may be shaped not only by their phrasing but also by the nature of the prompt being refused. H4 posits that ethical refusals are evaluated more positively when the underlying prompt is clearly sensitive or harmful. Relying on the OpenAI Moderation API, we classify prompts into flagged and unflagged categories to assess whether content sensitivity moderates the refusal penalty.

Table 7 presents win/loss/tie rates for ethical refusals across different moderation categories. Notably, the majority of prompts that elicited ethical refusals (70%) were not flagged by the Moderation API. These unflagged prompts often involved ethically ambiguous or low-stakes requests, such as lighthearted jokes or cheating in video gaming.

Comparing flagged and unflagged prompts, we find that ethical refusals to flagged content receive substantially more favorable evaluations. In user judgments, flagged prompts are associated with a 4 percentage point higher win rate and a roughly twofold increase in the win/loss ratio (0.28 vs. 0.13). These findings support H4: refusals are less penalized when the underlying request is clearly inappropriate or harmful.

In line with this interpretation, we observe comparatively high win rates for refusals triggered by prompts flagged as hate speech or illegal activity (15–17%), which typically involve clear ethical or legal boundaries. These results suggest that when users perceive a strong justification for the refusal, they are more likely to view it as acceptable or even appropriate.

However, the moderation categories in Table 7 should be interpreted with caution. While we find lower win rates for categories like sexual or violent content, this should not be taken to imply that these categories do not raise safety concerns. Rather, perceptions of such content may vary significantly based on platform norms, cultural expectations, or age-related considerations—such as the need to protect minors or comply with jurisdiction-specific content standards. It should also be noted that the sexual and violent categories, as defined by the OpenAI Moderation API, do not encompass illegal content; for example, the API includes a separate category for sexual content involving minors, which was not present in our dataset.

Table 7: Content Moderation Flagging and Win Rates for Refusal Ethical

		User Decision			
Moderation	**Count**	**Win**	**Loss**	**Tie**	**Win/Loss**
flagged	796	0.11	0.43	0.45	0.28
not flagged	1859	0.07	0.55	0.38	0.13
Moderation Categories					
harassment	98	0.11	0.45	0.44	0.24
hate	41	0.17	0.44	0.39	0.39
illicit	443	0.15	0.34	0.51	0.44
illicit violent	199	0.15	0.34	0.51	0.44
sexual	42	0.05	0.57	0.38	0.09
violence	234	0.05	0.57	0.38	0.09

5. Discussion

This study investigated how users respond to refusal behavior in large language models (LLMs) (H1), with a particular focus on the role of refusal type (H2a, H2b), refusal phrasing (H3a, H3b), and content sensitivity (H4). Drawing on a hand-labeled dataset of boundary-setting responses in Chatbot Arena, we found consistent evidence that users penalize refusal behavior—but with notable differences depending on how and when the refusal is issued.

5.1. The Refusal Penalty and User Satisfaction (H1 & H2)

Our results strongly support H1: refusal responses are systematically associated with lower user satisfaction, as indicated by reduced win rates in pairwise LLM response comparisons. The results align with prior work suggesting that users favor cooperative and responsive behavior from AI systems, and may interpret refusals as failures to engage rather than as signs of responsible design. While we also observe a performance penalty for disclaimers (to a lesser extent than outright refusals), a deeper conceptual discussion of disclaimer behavior is beyond the scope of this paper and addressed in a separate study (citation omitted for anonymity).

H2a and H2b further examined whether the type of refusal shapes user satisfaction. Consistent with H2a and in contrast to H2b, we find that ethical refusals—those grounded in safety or appropriateness concerns—were more strongly penalized than technical refusals that signal the model's limitations. Although both refusal types led to a decline in user preference, the drop was significantly steeper for ethical refusals. These findings suggest that technical refusals are seen as more understandable or acceptable, while ethical refusals may be perceived as overly cautious, evasive, or even patronizing—particularly when the model offers no opportunity for clarification or dialogue.

Taken together, these findings underscore a core tension in LLM deployment: the very behaviors that signal strong alignment—such as declining ethically risky prompts—can simultaneously undermine user satisfaction. This highlights a deeper challenge in alignment research, where users and developers may hold differing expectations of what constitutes "helpful" or "trustworthy" AI behavior. Importantly, we do not take a normative stance on how often models *should* refuse; rather, we aim to empirically document how such refusals are actually perceived in practice by users.

5.2. Refusal Design Strategy: Mitigating the Penalty (H3 & H4)

Despite the robust refusal penalty identified in our analysis—especially for ethical refusals—our findings also point to specific strategies that can help mitigate this dissatisfaction. Given that safety alignment remains a crucial design objective, the solution may not be to eliminate refusals entirely, but to optimize *how* and *when* they are delivered. Drawing on H3 and H4, we identify three actionable insights for designing more effective refusal strategies:

1. Avoid overly formulaic refusals.
Refusals that rely on generic templates—such as "*I am sorry I cannot help with that.*"—were consistently associated with lower win rates. These responses often fail to engage with the user's prompt and are perceived as unhelpful or dismissive. H3a highlights that refusals with *higher semantic similarity* to the user prompt are judged more favorably, suggesting that tailoring refusal language to reflect the user's intent improves perceived responsiveness.

2. Provide context, reasoning, or alternatives.
Similarly, longer responses that include explanations or offer constructive alternatives reduce dissatisfaction. As suggested by H3b, refusal length was positively associated with win rates for ethical refusals, indicating that users appreciate when the model "makes an effort" rather than shutting down the conversation. In contrast, for technical refusals, text length did not significantly improve ratings.

3. Calibrate moderation to avoid unnecessary refusals.
Our analysis of prompt sensitivity (H4) shows that users are more accepting of ethical refusals when the underlying prompt is clearly inappropriate or harmful (e.g., involving hate speech or illegal activities). However, the majority of ethical refusals were triggered by prompts not